COVER LETTER

I am writing to express my interest in the Software Developer position within the Robotics Department at Michigan Engineering. With over four years of professional experience in software development and my current pursuit of a Master's in Computer and Information Science at the University of Michigan - Dearborn, I am eager to contribute to Michigan Engineering's mission of building a people-first future through innovative technology.

In my previous roles as a Software Engineer, I have developed and maintained scalable web applications using Python, JavaScript, and RESTful APIs, ensuring high reliability and performance. My experience includes designing backend services, integrating database interactions using SQL, and collaborating with cross-functional teams to translate user requirements into effective software solutions. I have also contributed to debugging and application support, maintaining secure and compliant development practices throughout the software lifecycle.

Beyond technical proficiency, I take pride in my ability to communicate effectively with non-technical stakeholders and deliver practical tools that enhance user productivity skills that align well with this role's emphasis on collaboration and continuous improvement. I am especially drawn to Michigan Engineering's commitment to innovation and service, and I am motivated by the opportunity to develop software that supports the Robotics Department's academic and research excellence.

I am confident that my technical background, attention to detail, and collaborative mindset make me a strong fit for this position. I would welcome the opportunity to discuss how my skills and enthusiasm can contribute to your team's goals.

Thank you for considering my application. I am looking forward to hearing from you.

Sincerely,

Aayush Chopra

AAYUSH CHOPRA

9472759982 | aayushh@umich.edu | https://www.linkedin.com/in/aachopra/ | https://github.com/ayushchopra20

EDUCATION

Master of Science in Computer and Information Science **Sept 2024 - April 2026**
University of Michigan – Dearborn

Bachelor of Technology in Computer Science Engineering **July 2016 - June 2019**
Kurukshetra University, Kurukshetra, India

EXPERIENCE

University of Michigan – Dearborn **May 2025 - Present**
Volunteer, Research Assistant

- Developed real-time multi-sensor systems integrating thermal imaging and biosensors to analyze physiological signals with high accuracy.
- Built automated image processing pipelines using OpenCV and Python for thermal data preprocessing and enhancement.
- Applied machine learning models using Scikit-learn for signal classification, optimizing accuracy and reliability.
- Enhanced system performance and reproducibility by refining sensor fusion algorithms and improving data pipeline efficiency.

Ongraph Technology Pvt. Ltd. Noida, India **Jan 2020 - Aug 2024**
Software Engineer

- Planned and coordinated software releases across multiple repositories, improving delivery consistency and reducing rollback incidents by 25%.
- Designed CI/CD pipelines with Jenkins, GitHub Actions, and Docker, cutting deployment time by 40% and ensuring smooth integration with QA and DevOps workflows.
- Developed and maintained REST/gRPC APIs and microservices, enhancing scalability and enabling faster iteration cycles.
- Integrated third-party APIs (SAML SSO, Health Hero) and improved system interoperability, reducing API latency by 40%.
- Created automated monitoring scripts and build verification processes to support release readiness and post-release validation.
- Coordinated sprint planning, retrospectives, and release retros, boosting test coverage by 35% through improved collaboration between QA and development teams.
- Conducted risk assessments for production releases and collaborated on incident response strategies to minimize downtime.
- Architected optimized data models in PostgreSQL and MongoDB, reducing data retrieval time by 90%.

PROJECTS

DriveGuard – Smart Road Hazard Detection System | Python, OpenCV, Google Maps API, Free-Wili device, Cloud Integration

- Built an intelligent pothole and road hazard detection system using computer vision and sensor data for connected vehicles.
- Utilized Google Maps API to visualize detected hazards for municipal dashboards, enabling authorities to prioritize repairs.

Real-Time Crowd Behavior Analysis for Risk Mitigation | Python, OpenCV, TensorFlow, YOLO, Real-Time Video Processing

- Implemented real-time crowd analysis on edge devices (Raspberry Pi), reducing processing latency by 40%.
- Developed a basic alerting system that detects and flags crowd density spikes exceeding 75% of safe occupancy levels.

IoT-Based Real-Time Monitoring System | AWS IoT Core, Lambda, API Gateway, EC2, SNS, MQTT Broker, Raspberry Pi, Python

- Achieved a 30% reduction in system latency by optimizing MQTT communication and leveraging AWS IoT Core for efficient device-to-cloud data transmission.
- Leveraged serverless AWS services (Lambda, API Gateway) to minimize infrastructure costs, reducing operational overhead by 20% compared to traditional server-based solutions.

TECHNICAL SKILLS

- Programming Languages: Python, JavaScript (React), Ruby on Rails, C/C++, MATLAB
- Backend Development: REST/gRPC APIs, Microservices, Release Automation, CI/CD
- Database & Storage: PostgreSQL, MongoDB, MySQL
- Cloud & DevOps: AWS, GCP, Docker, Kubernetes, Jenkins, GitHub Actions, OpenShift, Ansible
- Tools And Technologies: Jira, Git, Postman, Linux, Agile/Scrum
- Methodologies: Infrastructure-as-Code, Configuration-as-Code, Agile Development, Release Management

ACADEMIC / EXTRA CURRICULAR ACTIVITES

- Completed Vector Embedded Software Training (5 days) - exposure to automotive software standards & practices.
- Recognized as a “Super Node” for exemplary performance at Ongraph Technology (2022)